# Notes: Ginzinger, Li, Niessen-Ruenzi & Wang (SSRN, 2026)
## The Motherhood Penalty in the Mutual Fund Industry

# Contents

# 1 Big picture

- **Question.** Does childbirth generate persistent career penalties for women in a high-skill finance occupation where output is unusually observable?
- **Setting.** Chinese mutual fund managers from 2006 to 2024. Maternity leaves longer than 30 days must be publicly disclosed to investors and regulators, allowing the authors to identify pregnancy, maternity leave, return dates, and career paths.
- **Why this setting is powerful.** Fund management has high-frequency measures of performance, flows, assets under management, portfolio responsibility, leadership roles, corporate site visits, and trading behavior. This lets the paper test whether the career penalty is proportional to observable productivity changes.
- **Main design.** Each new mother is compared to her pre-birth co-managers on the same fund(s). This comparison holds constant fund-level shocks, the investment mandate, fund company environment, and many unobserved career conditions.
- **Main career result.** Relative to pre-birth co-managers, new mothers are 25.5 percentage points less likely to remain active in the mutual fund industry. Relative to the average participation rate of 60.8%, this is about a 42% decline.
- **Main intensive-margin result.** Conditional career responsibility also falls: leadership positions fall by about 0.102 roles, normalized assets under management fall by 1.222 units, management responsibilities fall by 0.849, single-manager mandates fall by 0.201, and active-management mandates fall by 0.269.
- **Performance result.** There is a temporary performance decline during pregnancy for single-managed funds, but returns rebound after maternity leave. Team-managed funds show weaker or offsetting performance effects.
- **Investor response result.** Investor flows and institutional ownership do not systematically decline around childbirth. Investors do not appear to punish funds managed by pregnant or recently returned managers.
- **Effort result.** Managers reduce travel-intensive on-site company visits during pregnancy and after return, but online visits and total information acquisition are much more stable. Trading behavior is broadly stable.
- **Heterogeneity result.** Strong pre-birth performers do not receive a meaningfully smaller motherhood penalty. Performance during pregnancy and investor reactions also do not predict future career penalties.
- **Central takeaway.** The penalty is not well explained by persistent productivity deterioration, investor punishment, or negative performance selection. The most plausible margin is post-birth retention and allocation of career opportunities - who gets leadership roles, assets, and responsibility after leave.

The paper is best read as a "hard test" of performance-based explanations for motherhood penalties. If measurable productivity were the main reason mothers lose career ground, the mutual fund industry should reveal it: performance is public, pay-relevant, and observed frequently. The paper instead finds a mismatch: career consequences are large and persistent, while observable performance effects are small, temporary, and not predictive of career penalties. That mismatch is the paper's core insight.

# 2 Data and measurement

## 2.1 Institutional background

- The Chinese mutual fund industry is large and rapidly growing, with total assets under management of RMB 32.8 trillion by the end of 2024.
- The CSRC requires fund companies to disclose fund manager absences exceeding 30 days. Because statutory maternity leave exceeds this threshold, maternity leaves become observable.
- Maternity leave typically begins 15 days before the expected birth date. The authors infer approximate pregnancy timing as nine months before that point.
- Return announcements are observed for most maternity leaves, allowing the authors to distinguish returners from those who exit the industry.

**Interpretation:** The disclosure rule is the identification backbone of the paper. In most labor settings, childbirth timing, leave timing, career opportunities, and output are observed imperfectly. Here, the regulatory disclosure system turns an otherwise private event into a dated career event.

## 2.2 Sample construction

- The full fund-manager panel comes from the RESSET Survivor-Bias-Free Chinese Mutual Fund database.
- The fund-level sample includes 861,532 fund-month observations.
- The manager-level sample includes 327,928 manager-month observations.
- The authors identify 280 unique managers who take maternity leave by reading temporary departure disclosures.
- They drop managers with multiple maternity leaves to focus on first observed childbirth-related career interruption.
- The final maternity-leave sample contains 243 unique managers.
- No paternity-leave cases are observed in the sample period.

**Interpretation:** The sample restriction matters. Multiple births could compound career effects and complicate event timing. Restricting to first observed maternity leave gives a cleaner event-study interpretation.

## 2.3 Career outcomes

The paper studies both extensive and intensive margins:

- **Industry participation:** an indicator equal to one if a manager is responsible for at least one fund in a month. Once a manager exits the industry, she is coded as a nonparticipant for later months.
- **Leadership positions:** roles such as board director, investment strategy committee membership, executive roles, chief economist, party leadership, or other management positions.
- **Management responsibilities:** the number of funds a manager oversees, adjusted for co-management.
- **Single management:** whether the manager is solely responsible for at least one fund.
- **Active management:** whether the manager manages at least one actively managed fund.
- **Normalized AUM:** assets under management allocated to the manager after adjusting for team management and fund type.

A useful way to write normalized AUM is

$$\text{NormAUM}_{m,t} = \sum_{f \in \mathcal{F}_{m,t}} \frac{AUM_{f,t}/N_{f,t}}{\overline{AUM}_{type(f),t}}, \tag{1}$$

where $N_{f,t}$ is the number of co-managers on fund $f$ and $\overline{AUM}_{type(f),t}$ is the average AUM for funds of that type.

**Interpretation:** Normalized AUM is especially important because it is close to pay-relevant responsibility. Losing AUM is not just losing a statistic; it likely means losing prestige, compensation, future promotion opportunities, and organizational influence.

## 2.4 Performance, flows, and effort measures

- Monthly raw returns are observed for funds.
- One-factor abnormal returns use a combined equity-bond factor model estimated over rolling daily-return windows:

  $$RET_{fmd} - r_{fd} = \alpha_{fm} + \beta^{MKT}_{fm} MKT_d + \beta^{B}_{fm} B_d + \varepsilon_{fmd}. \tag{2}$$

  Daily abnormal returns are

  $$AR_{fmd} = (RET_{fmd} - r_{fd}) - \widehat{\beta}^{MKT}_{fm} MKT_d - \widehat{\beta}^{B}_{fm} B_d. \tag{3}$$

- Three-factor abnormal returns combine China-specific Fama-French equity factors with a bond market factor.
- Fund flows are measured as percentage changes in total net assets net of fund returns.
- Effort is proxied by corporate site visits, separated into on-site visits and online visits.
- Trading behavior is measured by total risk and non-market variation in returns.

**Interpretation:** The paper uses a layered test of the productivity explanation. It asks not only whether returns decline, but also whether investors respond, whether effort falls, whether trading changes, whether weaker managers sort into motherhood, and whether stronger managers avoid the penalty. This is why the negative conclusion about performance-based explanations is strong.

# 3 Models and empirical specifications

## 3.1 Event-study specification for career outcomes

For manager $m$ and year $t$, the baseline event-study model is

$$Y_{mt} = \sum_{\tau=-3,\tau\neq-1}^{3} \beta_\tau \mathbf{1}\{t \text{ is } \tau \text{ years from childbirth}\} + \gamma Treated_m + \Phi_{ct} + \Psi_{mt} + \varepsilon_{mt}. \tag{4}$$

- $Y_{mt}$ is a career outcome: participation, leadership, normalized AUM, responsibilities, single management, or active management.
- $Treated_m = 1$ for managers who experience childbirth during the sample.
- $\Phi_{ct}$ are year-by-entry-cohort fixed effects.
- $\Psi_{mt}$ are manager-age fixed effects.
- The omitted event year is $t = -1$, the year before childbirth.

**Interpretation:** The event study asks whether treated managers were already on a different career path before motherhood and how sharply outcomes change after childbirth. The clean pre-trends are important because they reduce concern that women were already exiting or losing responsibility before pregnancy.

## 3.2 Pooled difference-in-differences career specification

The pooled version summarizes the pre-post difference:

$$Y_{mt} = \alpha + \beta Motherhood_{mt} + \gamma Treated_m + \Phi_{ct} + \Psi_{mt} + u_{mt}, \tag{5}$$

where $Motherhood_{mt}$ indicates the post-childbirth period for managers who become mothers.

**Interpretation:** The coefficient $\beta$ is the average post-birth career penalty relative to the comparison group. The paper estimates it using three comparison groups: all co-managers, female co-managers, and all non-mother female managers.

## 3.3 Performance around childbirth

At the fund-manager-month level, the paper estimates

$$\begin{aligned} Performance_{fmt} = \beta_0 &+ \beta_1 Pregnancy_{mt} + \beta_2 Post0to6_{mt} + \beta_3 Post6to12_{mt} && (6)\\ &+ \beta_4 Post12to36_{mt} + \gamma Treated_m + \beta_5 X_{fmt} + \Phi_{fmt} + \varepsilon_{fmt}. && (7)\end{aligned}$$

The event windows are:

- **Before:** 36 months before pregnancy.
- **Pregnancy:** month 0 to month 9 before maternity leave starts.
- **Post0to6:** first 6 months after return.
- **Post6to12:** months 6 to 12 after return.
- **Post12to36:** months 12 to 36 after return.

**Interpretation:** The split is economically meaningful. Pregnancy may generate temporary constraints on attention and travel; the first six months after return may involve infant care and lactation breaks; the first three years after birth may be the highest childcare-intensity period.

### 3.4 Team-management interaction

The performance tables add interactions with $TeamFund_f$:

$$Performance_{fmt} = \cdots + \theta_1 (Pregnancy_{mt} \times Treated_m \times TeamFund_f) + \cdots . \tag{8}$$

**Interpretation:** Team management is a built-in organizational insurance mechanism. If teams absorb temporary attention shocks, performance should fall more for single-manager funds than team-managed funds. That is what the results suggest.

### 3.5 Effort and trading specifications

For visits or trading outcomes, the paper estimates analogous event-window regressions:

$$Effort_{mt} = \alpha + \beta_1 Pregnancy_{mt} + \beta_2 Post0to6_{mt} + \beta_3 Post6to12_{mt} + \beta_4 Post12to36_{mt} + \Gamma X_{mt} + \lambda_t + e_{mt}. \tag{9}$$

**Interpretation:** The effort test distinguishes between lower total effort and changed effort composition. A shift from on-site to online visits is not the same as broad withdrawal from work.

### 3.6 Performance heterogeneity in the motherhood penalty

The paper tests whether high performers suffer less:

$$Y_{mt} = \alpha + \beta Motherhood_{mt} + \theta (Motherhood_{mt} \times TopPerformer_m) + \delta TopPerformer_m + FE + e_{mt}. \tag{10}$$

**Interpretation:** If firms were rationally reallocating responsibility based on performance, high pre-birth performers should be protected. The evidence shows that they are not. This is one of the paper's most important mechanism tests.

# 4 Identification, interpretation, and limits

## 4.1 Why the co-manager design is strong

- Co-managers share the same fund, organization, incentives, investment mandate, client base, and many unobservable shocks.
- The design compares managers who are already professionally close substitutes before childbirth.
- This is stronger than comparing mothers to unrelated workers in the same industry because fund-level conditions are held approximately fixed.

The key identification insight is that the authors do not compare new mothers to "all other managers" first. They compare them to people with whom they jointly ran the same fund immediately before the event. The co-manager is a natural counterfactual because that person was exposed to the same fund environment but did not have the childbirth event.

## 4.2 What the design can and cannot identify

The design can identify whether career outcomes diverge around childbirth and whether the divergence is aligned with performance, flows, and effort. It cannot cleanly separate:

- **supply-side mechanisms:** mothers choosing fewer responsibilities, declining roles, leaving the industry, or moving to less demanding work;
- **demand-side mechanisms:** firms reallocating responsibilities, promotions, leadership positions, and AUM away from mothers;
- **outside constraints:** childcare availability, family support, commuting, or health conditions that are not recorded in the fund data.

**Interpretation:** The paper's strongest claim is not "employers discriminate." The strongest claim is narrower and more defensible: large career penalties are hard to explain by observable performance, investor flows, measured effort, fertility timing, or selective return.

## 4.3 Why performance does not explain the career penalty

- Single-managed funds experience a negative performance effect during pregnancy.
- The effect is temporary and does not persist after maternity leave.
- Team-managed funds show weaker or offsetting effects.
- Investors do not reduce flows or institutional holdings around childbirth.
- Top pre-birth performers do not avoid the penalty.
- Pregnancy-period performance does not predict subsequent exit or responsibility loss.

This creates a "magnitude mismatch." Career penalties are persistent and large; performance effects are temporary and modest. The asymmetry suggests that the post-birth career process is governed less by realized performance and more by responsibility allocation, retention frictions, or beliefs about future availability.

## 4.4 Potential mechanisms consistent with the results

The evidence is consistent with several mechanisms, none of which can be separately proven by the data:

- **Responsibility reallocation during leave.** Replacement or co-managers may inherit mandates that are not fully returned after maternity leave.

- **Beliefs about future commitment.** Employers may infer lower long-run availability from childbirth even when realized performance does not support that inference.
- **Self-selection into different roles.** Mothers may prefer fewer travel-intensive or leadership responsibilities after childbirth.
- **Organizational inertia.** Temporary leave arrangements can become permanent because portfolios, clients, and internal leadership pipelines adjust during the leave period.
- **Implicit tournament effects.** In high-paying finance jobs, brief interruptions may have outsized effects if promotion tournaments are steep and timing-sensitive.

**Interpretation:** The paper's data cannot distinguish these channels because it does not observe internal offer sets or preferences. But the data do rule out a simple story in which firms merely respond to sustained lower investment skill.

# 5 Tables and figures

## 5.1 Table 1 + Figure 1: Baseline sample and gender composition

**Read:**

- Table 1 reports 861,532 fund-month and 327,928 manager-month observations.
- Female managers are 22.8% of the manager-month sample.
- Average industry participation is 60.7%; average normalized AUM is 3.183.
- Figure 1 shows that female fund managers grow from 31 in 2006 to 1,063 in 2024.
- The female share rises from 9.1% to 26.3%, but women remain underrepresented relative to population shares.

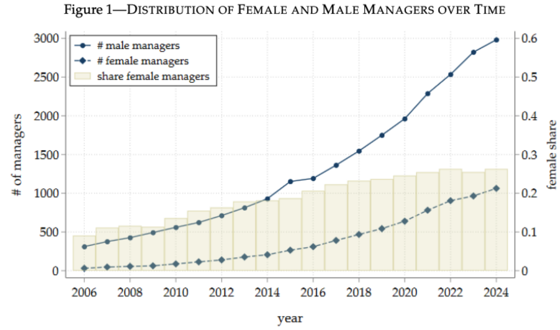
**Economic message:** The industry becomes more gender diverse over time, but the baseline data already show that women remain a minority in fund management. The motherhood event is therefore studied in an elite and still male-dominated occupation.

Table 1—FUND AND MANAGER SUMMARY STATISTICS

| | Mean | STD | Median | P25 | P75 |
|---|---|---|---|---|---|
| **Panel A: Fund-level summary statistics** (N = 861,532) | | | | | |
| Fund size (in millions) | 1497.291 | 2590.389 | 492.924 | 132.218 | 1596.055 |
| Fund age (in years) | 4.048 | 3.716 | 3.000 | 1.333 | 5.667 |
| Fund flows | 0.076 | 0.803 | -0.021 | -0.115 | 0.007 |
| Raw return (in %) | 0.407 | 4.856 | 0.252 | -1.197 | 1.554 |
| 1F Abnormal return (in %) | 0.092 | 2.849 | 0.073 | -0.844 | 0.927 |
| 3F Abnormal return (in %) | 0.177 | 2.563 | 0.093 | -0.682 | 0.948 |
| Total risk (in %) | 3.632 | 3.093 | 3.469 | 0.522 | 5.886 |
| Idiosyncratic risk (in %) | 1.972 | 1.792 | 1.585 | 0.342 | 3.081 |
| $\beta_{Market}$ | 0.738 | 3.340 | 0.558 | -0.177 | 1.684 |
| Number of managers | 1.307 | 0.521 | 1.000 | 1.000 | 2.000 |
| Non-market variation | 0.363 | 0.243 | 0.304 | 0.168 | 0.514 |
| **Panel B: Manager-level summary statistics** (N = 327,928) | | | | | |
| Manager age (in years) | 36.231 | 4.892 | 36.000 | 33.000 | 39.000 |
| Manager fund tenure (in years) | 1.757 | 1.792 | 1.250 | 0.500 | 2.417 |
| Manager is female | 0.228 | 0.420 | 0.000 | 0.000 | 0.000 |
| PhD as highest degree | 0.119 | 0.324 | 0.000 | 0.000 | 0.000 |
| Master as highest degree | 0.844 | 0.362 | 1.000 | 1.000 | 1.000 |
| Bachelor as highest degree | 0.037 | 0.188 | 0.000 | 0.000 | 0.000 |
| Industry participation | 0.607 | 0.489 | 1.000 | 0.000 | 1.000 |
| # Leadership positions | 0.392 | 0.818 | 0.000 | 0.000 | 0.000 |
| # (Co-) managed funds | 3.466 | 2.924 | 2.000 | 1.000 | 5.000 |
| # Management responsibilities | 2.654 | 2.429 | 2.000 | 1.000 | 3.500 |
| Normalized AUM | 3.183 | 5.369 | 1.127 | 0.259 | 3.686 |
| P(Single management) | 0.729 | 0.445 | 1.000 | 0.000 | 1.000 |
| P(Active management) | 0.952 | 0.213 | 1.000 | 1.000 | 1.000 |

**(a)** Table 1: fund and manager summary statistics

Figure 1—DISTRIBUTION OF FEMALE AND MALE MANAGERS OVER TIME




*Notes:* The figure illustrates the distribution of female and male managers over time. The solid line represents the total number of male managers, while the dashed line represents the total number of female managers. The bars indicate the share of female managers as a percentage of the total number of managers. Data are taken from the RESSET Survivor-Bias-Free Chinese Mutual Fund database. Our sample covers the time period from January 2006 to December 2024.

**(b)** Figure 1: female and male managers over time

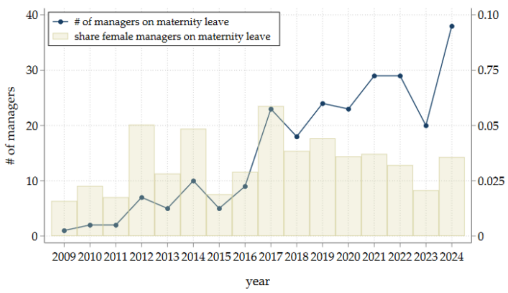
## 5.2 Figure 2 + Appendix Figure A2: Maternity events and age at leave

**Read:**

- Figure 2 shows annual maternity leaves rising over time, with the first observed case in 2009 and 38 cases in 2024.
- The average annual share of active female managers on maternity leave is about 3.14%.
- Appendix Figure A2 shows that maternity leave is concentrated around ages 30 to 37.
- The average maternity-leave age is about 33.9, slightly younger than the full manager average of 36.2.

**Economic message:** Maternity leave occurs relatively early in fund managers' careers, exactly when leadership pipelines and AUM responsibility may still be forming. That timing helps explain why a temporary leave can have persistent career consequences.

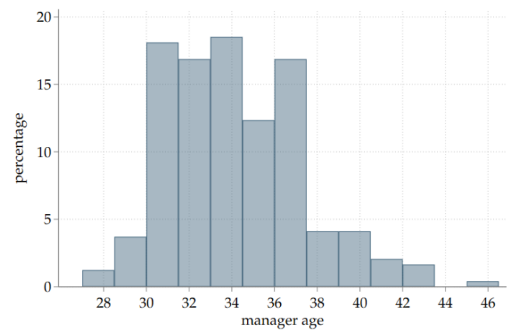
Figure 2—Maternity Leaves over Time




*Notes:* This figure shows the annual number and share of maternity leaves taken by female fund managers. The line plots the total number of managers who take a maternity leave in a given year, while the bars show the share of female managers who take a maternity leave relative to all active female managers. We do not observe any instances of paternity leave. The sample is drawn from the RESSET Survivor-Bias-Free Chinese Mutual Fund database and covers the time period January 2006 to December 2024. The figure begins in 2009, the first year in which we observe a maternity leave. Maternity leaves are identified by reviewing the departure disclosures and selecting cases that explicitly state maternity leave as reason.

**(a)** Figure 2: maternity leaves over time

Figure A2—Manager Age at Time of Maternity Leave




*Notes:* The figure displays the distribution of manager age at the time of maternity leave. Manager age is estimated based on education history and career entry into the mutual fund industry. A detailded description is provided in Appendix Table A1. The underlying data are taken from the RESSET Survivor-Bias-Free Chinese Mutual Fund database and cover the period from January 2006 to December 2024.

**(b)** Figure A2: manager age at maternity leave

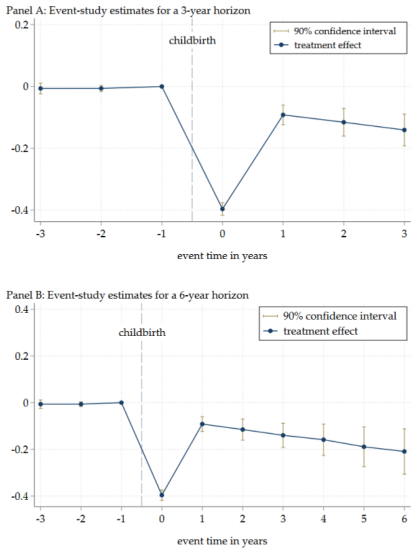
## 5.3 Figure 3 + Figure 4: Event-study evidence on career outcomes

**Read:**

- Figure 3 shows near-zero pre-birth differences in participation relative to co-managers.
- Participation drops sharply at childbirth and remains below the co-manager counterfactual several years later.
- Figure 4 shows intensive-margin losses in leadership and normalized AUM.
- Leadership and AUM losses remain visible after the immediate maternity-leave period.

**Economic message:** These figures are the visual core of the paper. They show that the motherhood penalty is not only temporary leave; it is a persistent divergence in career trajectory.

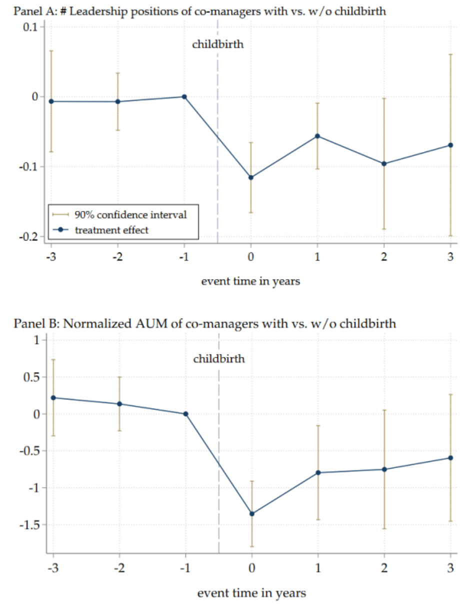
Figure 3—Extensive margin: The effect of motherhood on industry participation




*Notes*: The figure compares participation rates in the mutual fund industry between managers who become mothers and co-managers who do not. Participation is defined unconditionally on employment status: once a manager exits the mutual fund industry, she is coded as a non-participant for all subsequent months. Event time is defined relative to childbirth, with $t = 0$ denoting the year of childbirth. The coefficient for $t = -1$ is omitted so that all effects are estimated relative to the year preceding childbirth. All specifications control for cohort-by-year and manager-age fixed effects. The 90 percent confidence intervals are based on standard errors clustered at the manager and year levels. Panel B extends the event-study horizon to six years; because most childbirth events occur after 2017, later coefficients in the extended horizon are based on fewer observations.

**(a)** Figure 3: extensive-margin participation effect

Figure 4—Intensive margin: The effect of motherhood on leadership and AUM




*Notes*: The figure compares the career trajectories of co-managers with versus without childbirth in the mutual fund industry. The estimates represent event-study estimates of the effect of motherhood. Panel A displays the estimated impact on the number of leadership roles within fund companies, and Panel B shows the corresponding estimates for person-level normalized AUM. Event time is measured relative to childbirth, with $t = 0$ denoting the year of childbirth; the coefficient for $t = -1$ is omitted so that all effects are relative to the year preceding childbirth. All specifications include cohort-by-year and manager-age fixed effects. The 90 percent confidence intervals are based on standard errors clustered at the manager and year level.

**(b)** Figure 4: leadership and normalized AUM effects

## 5.4 Table 2: Pooled career penalties

**Read:**

- Relative to co-managers, motherhood reduces participation by 0.255 percentage points, leadership roles by 0.102, normalized AUM by 1.222, management responsibilities by 0.849, single-management probability by 0.201, and active-management probability by 0.269.
- The participation coefficient corresponds to about a 42% decline relative to the 60.8% participation mean.
- The leadership coefficient corresponds to about a 26% decline relative to the 0.39 leadership-role mean.
- The normalized AUM coefficient corresponds to about a 38% decline relative to the mean of 3.18.
- Similar patterns hold when the control group is restricted to female co-managers or all other non-mother female managers.

**Economic message:** This is not a one-dimensional penalty. Mothers lose industry attachment, internal influence, AUM, and more prestigious portfolio assignments. The breadth of the effect points to a broad responsibility-allocation process after childbirth.

Table 2—MOTHERHOOD AND CAREER OUTCOMES

| | Extensive margin: Industry participation | # Leadership positions | Normalized AUM | Intensive margin: # Management responsibilities | P(single management) | P(active management) |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| **Panel A: Co-manager control group** | | | | | | |
| Motherhood | -0.255*** | -0.102** | -1.222*** | -0.849*** | -0.201*** | -0.269*** |
| | (0.017) | (0.044) | (0.308) | (0.140) | (0.033) | (0.016) |
| N | 50,186 | 41,545 | 41,373 | 41,373 | 41,373 | 41,373 |
| $R^2$ | 0.150 | 0.236 | 0.169 | 0.261 | 0.191 | 0.128 |
| **Panel B: Female co-manager control group** | | | | | | |
| Motherhood | -0.311*** | -0.071** | -1.132*** | -0.905*** | -0.213*** | -0.302*** |
| | (0.021) | (0.030) | (0.371) | (0.172) | (0.035) | (0.018) |
| N | 28,239 | 23,939 | 23,888 | 23,888 | 23,888 | 23,888 |
| $R^2$ | 0.201 | 0.267 | 0.204 | 0.272 | 0.211 | 0.151 |
| **Panel C: Female control group** | | | | | | |
| Motherhood | -0.186*** | -0.068* | -0.866*** | -0.784*** | -0.185*** | -0.251*** |
| | (0.021) | (0.033) | (0.247) | (0.126) | (0.032) | (0.017) |
| N | 103,832 | 75,075 | 74,826 | 74,826 | 74,826 | 74,826 |
| $R^2$ | 0.242 | 0.234 | 0.115 | 0.201 | 0.152 | 0.067 |
| Year-by-Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Age FE | Yes | Yes | Yes | Yes | Yes | Yes |

*Notes*: This table reports pooled difference-in-differences estimates based on Equation (1), summarizing the average effect of motherhood on career outcomes. The coefficients capture changes in outcomes from the pre- to the post-childbirth period for managers who become mothers relative to the relevant comparison group. Columns (1) to (3) focus on three key career outcomes: industry participation, the number of leadership positions, and person-level normalized AUM. Columns (4) to (6) report results for effective management responsibilities and the probabilities of single and active management, capturing shifts in the scope and structure of portfolio oversight that should be correlated with fund managers' prestige and eventually their career advancement. The main comparison group consists of co-managers who jointly oversaw the same fund immediately prior to childbirth (Panel A). As robustness, we also estimate the specification using female co-managers only (Panel B) and all other female managers in the industry who do not become mothers (Panel C). All regressions include year-by-entry-cohort and manager-age fixed effects. Standard errors are clustered at the manager and year level.

Table 2: motherhood and career outcomes

## 5.5 Table 3 + Table 4: Performance around childbirth

**Read:**

- Table 3 shows that single-managed funds experience negative returns during pregnancy: raw returns and abnormal returns fall, with the largest estimates roughly 0.8 to 1.3 percentage points annualized as discussed in the paper.
- Post-maternity-leave coefficients are generally insignificant, indicating rebound after leave.
- Team-fund interactions are positive in pregnancy for some abnormal-return specifications, consistent with co-managers absorbing the temporary shock.
- Table 4 uses a matched sample based on fund type, style, size, and age and reaches the same broad conclusion: performance effects are not persistent.

**Economic message:** The performance results are not irrelevant - pregnancy can temporarily disrupt output, especially in single-manager settings. But the timing and persistence are wrong for explaining the career results. Career penalties remain after performance normalizes.

## 5.6 Table 5 + Appendix Figure A3: Effort and information acquisition

**Read:**

- Table 5 shows that on-site company visits fall during pregnancy by 0.080 visits per month, significant at the 1% level.
- The on-site reduction is large relative to the sample's average visit frequency.
- Online visits do not change significantly.
- Appendix Figure A3 shows that online visits rise sharply after 2020, so separating on-site and online information acquisition is important.

Table 3—Fund Performance around Childbirth

| | Raw Return (in %) | | 1F Return (in %) | | 3F Return (in %) | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Pregnancy ($0 \le t_P < 9$) × Treated | -0.221** | -0.107* | -0.150** | -0.081* | -0.106* | -0.067 |
| | (0.108) | (0.063) | (0.067) | (0.048) | (0.059) | (0.046) |
| Post Maternity Leave ($0 < t_{PM} \le 6$) × Treated | -0.037 | -0.054 | -0.052 | -0.054 | -0.022 | -0.051 |
| | (0.114) | (0.080) | (0.076) | (0.070) | (0.072) | (0.066) |
| Post Maternity Leave ($6 < t_{PM} \le 12$) × Treated | 0.112 | -0.021 | 0.038 | -0.047 | 0.016 | -0.042 |
| | (0.126) | (0.093) | (0.086) | (0.078) | (0.078) | (0.074) |
| Post Maternity Leave ($12 < t_{PM} \le 36$) × Treated | 0.099 | 0.054 | 0.043 | 0.026 | 0.025 | 0.006 |
| | (0.114) | (0.083) | (0.072) | (0.062) | (0.064) | (0.058) |
| Pregnancy ($0 \le t_P < 9$) × Treated × Team Fund | 0.164 | 0.090 | 0.127 | 0.091 | 0.136** | 0.115* |
| | (0.103) | (0.086) | (0.083) | (0.069) | (0.069) | (0.061) |
| Post Maternity Leave ($0 < t_{PM} \le 6$) × Treated × Team Fund | 0.054 | 0.037 | 0.009 | -0.048 | 0.014 | -0.019 |
| | (0.161) | (0.128) | (0.119) | (0.104) | (0.107) | (0.098) |
| Post Maternity Leave ($6 < t_{PM} \le 12$) × Treated × Team Fund | -0.033 | 0.032 | -0.062 | 0.011 | -0.005 | 0.046 |
| | (0.159) | (0.140) | (0.113) | (0.113) | (0.099) | (0.102) |
| Post Maternity Leave ($12 < t_{PM} \le 36$) × Treated × Team Fund | -0.032 | 0.090 | 0.036 | 0.126 | 0.054 | 0.114 |
| | (0.109) | (0.098) | (0.101) | (0.095) | (0.080) | (0.080) |
| Treated | 0.001 | 0.009 | 0.016 | 0.023 | -0.001 | 0.010 |
| | (0.063) | (0.038) | (0.045) | (0.036) | (0.039) | (0.033) |
| Team Fund | -0.059 | -0.051 | -0.058 | -0.050 | -0.079 | -0.071 |
| | (0.083) | (0.077) | (0.069) | (0.064) | (0.057) | (0.054) |
| Log(Fund Size) | 0.050*** | 0.022** | 0.042*** | 0.024*** | 0.038*** | 0.024*** |
| | (0.013) | (0.009) | (0.008) | (0.007) | (0.008) | (0.007) |
| Fund Age | 0.005 | 0.004 | -0.006 | -0.004 | 0.001 | 0.003 |
| | (0.010) | (0.010) | (0.005) | (0.005) | (0.005) | (0.005) |
| Manager Age | -0.004 | -0.000 | -0.004 | 0.000 | -0.004 | -0.001 |
| | (0.004) | (0.003) | (0.003) | (0.002) | (0.002) | (0.002) |
| PhD as highest degree | 0.007 | 0.073 | 0.017 | 0.052 | 0.010 | 0.039 |
| | (0.057) | (0.056) | (0.063) | (0.060) | (0.056) | (0.054) |
| Master as highest degree | -0.021 | 0.067* | 0.013 | 0.071* | 0.033 | 0.068 |
| | (0.052) | (0.035) | (0.046) | (0.042) | (0.045) | (0.041) |
| Month FE | Yes | No | Yes | No | Yes | No |
| Fund Type FE | Yes | No | Yes | No | Yes | No |
| Month × Fund Type FE | No | Yes | No | Yes | No | Yes |
| N | 107994 | 107970 | 107356 | 107332 | 107071 | 107047 |
| $R^2$ | 0.331 | 0.549 | 0.138 | 0.229 | 0.105 | 0.172 |

Table 4—Fund Performance around Childbirth: Matched Sample Analysis

| | Raw Return (in %) | | 1F Return (in %) | | 3F Return (in %) | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Pregnancy ($0 \le t_P < 9$) × Treated | -0.051 | -0.052 | -0.059 | -0.063 | -0.096* | -0.108** |
| | (0.055) | (0.061) | (0.056) | (0.049) | (0.049) | (0.047) |
| Post Maternity Leave ($0 < t_{PM} \le 6$) × Treated | -0.152* | -0.209** | -0.178* | -0.170** | -0.175** | -0.191** |
| | (0.087) | (0.093) | (0.092) | (0.086) | (0.078) | (0.077) |
| Post Maternity Leave ($6 < t_{PM} \le 12$) × Treated | 0.113 | 0.059 | -0.004 | -0.020 | -0.073 | -0.080 |
| | (0.137) | (0.127) | (0.116) | (0.108) | (0.104) | (0.101) |
| Post Maternity Leave ($12 < t_{PM} \le 36$) × Treated | 0.028 | 0.015 | -0.040 | -0.024 | -0.044 | -0.048 |
| | (0.098) | (0.081) | (0.088) | (0.073) | (0.071) | (0.063) |
| Pregnancy ($0 \le t_P < 9$) × Treated × Team Fund | 0.042 | 0.099 | 0.082 | 0.113 | 0.149* | 0.168** |
| | (0.095) | (0.087) | (0.093) | (0.085) | (0.080) | (0.077) |
| Post Maternity Leave ($0 < t_{PM} \le 6$) × Treated × Team Fund | 0.225 | 0.230 | 0.238* | 0.204* | 0.244** | 0.236** |
| | (0.152) | (0.140) | (0.133) | (0.120) | (0.121) | (0.115) |
| Post Maternity Leave ($6 < t_{PM} \le 12$) × Treated × Team Fund | -0.269 | -0.092 | -0.035 | 0.026 | 0.067 | 0.109 |
| | (0.194) | (0.180) | (0.158) | (0.156) | (0.145) | (0.142) |
| Post Maternity Leave ($12 < t_{PM} \le 36$) × Treated × Team Fund | 0.026 | 0.138 | 0.070 | 0.125 | 0.086 | 0.117 |
| | (0.148) | (0.134) | (0.133) | (0.126) | (0.104) | (0.102) |
| Team Fund | -0.053 | -0.038 | -0.013 | -0.013 | 0.024 | 0.011 |
| | (0.058) | (0.055) | (0.044) | (0.040) | (0.039) | (0.037) |
| Treated × Team Fund | 0.023 | -0.016 | -0.003 | -0.011 | -0.052 | -0.052 |
| | (0.066) | (0.065) | (0.068) | (0.065) | (0.060) | (0.060) |
| Treated | -0.015 | 0.006 | 0.014 | 0.018 | 0.025 | 0.034 |
| | (0.040) | (0.041) | (0.040) | (0.037) | (0.033) | (0.033) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Month FE | Yes | No | Yes | No | Yes | No |
| Fund type FE | Yes | No | Yes | No | Yes | No |
| Month × Fund type FE | No | Yes | No | Yes | No | Yes |
| Observations | 63945 | 63885 | 63479 | 63419 | 63265 | 63205 |
| $R^2$ | 0.309 | 0.529 | 0.104 | 0.216 | 0.086 | 0.174 |

*Notes*: This table reports coefficients from matched-sample difference-in-differences regressions of fund performance around childbirth, allowing the effects to differ between single-managed and team-managed funds. Fund performance is expressed in percent and measured using monthly raw returns and two abnormal return metrics. Each treated fund is matched to two control funds based on fund type, investment style, fund size, and fund age, using nearest-neighbor

**Economic message:** The effort evidence is nuanced. Managers do not simply stop acquiring information; they substitute away from travel-intensive information acquisition. This looks like a work-organization constraint rather than a collapse in effort or investment ability.

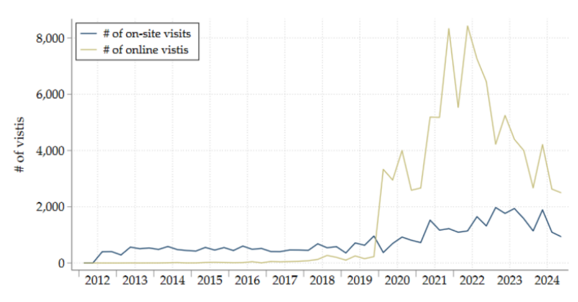
Table 5—Frequency of Company Visits around Childbirth

| | Total Visits | On-Site Visits | Online Visits |
|---|---|---|---|
| | (1) | (2) | (3) |
| Pregnancy ($0 \le t_P < 9$) × Treated | -0.105 | -0.080*** | -0.024 |
| | (0.070) | (0.021) | (0.064) |
| Post Maternity Leave ($0 < t_{PM} \le 6$) × Treated | -0.157* | -0.051** | -0.106 |
| | (0.087) | (0.025) | (0.082) |
| Post Maternity Leave ($6 < t_{PM} \le 12$) × Treated | -0.050 | -0.032 | -0.018 |
| | (0.105) | (0.025) | (0.098) |
| Post Maternity Leave ($12 < t_{PM} \le 36$) × Treated | -0.014 | -0.040* | 0.025 |
| | (0.092) | (0.022) | (0.083) |
| Treated | 0.001 | 0.010 | -0.009 |
| | (0.072) | (0.023) | (0.059) |
| Manager Age | -0.004 | -0.005*** | 0.001 |
| | (0.008) | (0.002) | (0.007) |
| PhD as highest degree | 0.004 | 0.051* | -0.048 |
| | (0.100) | (0.028) | (0.081) |
| Master as highest degree | 0.330*** | 0.107*** | 0.223*** |
| | (0.080) | (0.021) | (0.066) |
| Log (Fund Size, average) | -0.016 | 0.005 | -0.021 |
| | (0.020) | (0.005) | (0.016) |
| Fund Age (average) | 0.026** | 0.009*** | 0.018** |
| | (0.011) | (0.003) | (0.009) |
| Equity Fund (share) | 0.847*** | 0.254*** | 0.593*** |
| | (0.171) | (0.043) | (0.136) |
| Hybrid Fund (share) | 0.733*** | 0.218*** | 0.515*** |
| | (0.082) | (0.018) | (0.071) |
| Month FE | Yes | Yes | Yes |
| N | 31165 | 31165 | 31165 |
| $R^2$ | 0.088 | 0.021 | 0.094 |

*Notes*: This table reports coefficient estimates from pooled panel regressions of managers' corporate site visits around childbirth using untreated female managers as a control group. The sample includes all treated female managers (i.e., managers who experience a childbirth event) and all other female managers who are observed in the site-visit data at least once. The dependent variables are monthly visit counts at the manager level. *On-site visits* refer to physical meetings at portfolio companies; *online visits* include virtual interactions conducted by phone, email, or online platforms. The key explanatory variables are interaction terms between a *Treated* indicator and event-period indicators for *Pregnancy* ($0 \le t_P < 9$) and three mutually exclusive post-maternity-leave periods: *Post Maternity Leave* ($0 < t_{PM} \le 6$), *Post Maternity Leave* ($6 < t_{PM} \le 12$), and *Post Maternity Leave* ($12 < t_{PM} \le 36$). *Before* (the 36 months prior to pregnancy) is the omitted category. Control variables are described in Appendix Table A1. All specifications include month fixed effects. Standard errors are clustered at the manager and month level. $^*$, $^{**}$, and $^{***}$ denote significance at the 10%, 5%, and 1% levels, respectively.

**(a)** Table 5: company visits around childbirth

Figure A3—Number of Site Visits over Time




*Notes:* This figure plots the number of total on-site and online portfolio company visits over time, respectively. The information on company visits is available from July 2012 for firms listed on the Shenzhen Stock Exchange. An event is categorized as an online visit if the meeting is held via a digital method, e.g., email, phone, or online meetings.

**(b)** Figure A3: on-site and online visits over time

## 5.7 Table 6 + Appendix Table A13: Does performance explain the penalty?

**Read:**

- Table 6 interacts motherhood with pre-birth top performance.

- The main motherhood coefficients remain negative and significant across career outcomes.
- The interaction terms $Motherhood \times TopPerformer$ are small and statistically insignificant.
- Appendix Table A13 shows that pregnancy-period performance and investor-response variation do not explain later career outcomes.

**Economic message:** These are the paper's cleanest mechanism tables. If career penalties were performance-based, high-performing mothers or mothers with better pregnancy-period performance should be protected. They are not.

Table 6—THE MOTHERHOOD PENALTY IN THE MUTUAL FUND INDUSTRY: HETEROGENEITY BY PERFORMANCE

| | **Extensive margin:** Industry participation | # Leadership positions | Normalized AUM | **Intensive margin:** # Management responsibilities | P(single management) | P(active management) |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| **Panel A: Co-manager control group** | | | | | | |
| Motherhood | -0.281*** | -0.111** | -1.260** | -0.942*** | -0.199*** | -0.258*** |
| | (0.024) | (0.046) | (0.465) | (0.190) | (0.035) | (0.013) |
| Motherhood × Top Performer | 0.069 | 0.011 | 0.241 | 0.298 | -0.003 | -0.036 |
| | (0.057) | (0.082) | (0.655) | (0.348) | (0.061) | (0.035) |
| Top Performer | 0.018 | -0.108 | 0.966 | 0.279 | 0.025 | -0.044** |
| | (0.013) | (0.070) | (0.561) | (0.272) | (0.046) | (0.017) |
| N | 49947 | 41545 | 41373 | 41373 | 41373 | 41373 |
| $R^2$ | 0.156 | 0.237 | 0.263 | 0.169 | 0.191 | 0.131 |
| **Panel B: Female co-manager control group** | | | | | | |
| Motherhood | -0.330*** | -0.078* | -1.113** | -0.941*** | -0.202*** | -0.287*** |
| | (0.025) | (0.038) | (0.505) | (0.217) | (0.037) | (0.016) |
| Motherhood × Top Performer | 0.053 | 0.012 | 0.191 | 0.184 | -0.027 | -0.055 |
| | (0.053) | (0.066) | (0.699) | (0.368) | (0.059) | (0.033) |
| Top Performer | 0.013 | -0.053 | 1.171** | 0.354 | 0.036 | -0.048** |
| | (0.014) | (0.054) | (0.546) | (0.270) | (0.045) | (0.019) |
| N | 28000 | 23939 | 23888 | 23888 | 23888 | 23888 |
| $R^2$ | 0.244 | 0.268 | 0.274 | 0.205 | 0.211 | 0.156 |
| **Panel C: Female control group** | | | | | | |
| Motherhood | -0.205*** | -0.064* | -1.006** | -0.881*** | -0.187*** | -0.239*** |
| | (0.028) | (0.037) | (0.404) | (0.191) | (0.034) | (0.013) |
| Motherhood × Top Performer | 0.054 | -0.016 | 0.474 | 0.300 | 0.008 | -0.038 |
| | (0.058) | (0.071) | (0.641) | (0.350) | (0.057) | (0.033) |
| Top Performer | 0.050** | -0.057 | 0.991* | 0.373 | 0.012 | -0.036** |
| | (0.023) | (0.060) | (0.547) | (0.264) | (0.045) | (0.017) |
| N | 103593 | 75075 | 74826 | 74826 | 74826 | 74826 |
| $R^2$ | 0.244 | 0.235 | 0.119 | 0.205 | 0.155 | 0.092 |
| Year-by-Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Age FE | Yes | Yes | Yes | Yes | Yes | Yes |

*Notes*: This table reports pooled difference-in-differences estimates based on Equation (1), allowing the motherhood penalty to differ between top and non-top performers. The indicator *Motherhood* captures the change in outcomes from the pre- to the post-childbirth period for managers who become mothers relative to the relevant comparison group. *Top Performer* identifies treated managers whose cumulative abnormal returns (FF3-adjusted) in the pre-leave period place them in the top tercile of the treated-manager distribution. The interaction term *Motherhood × Top Performer* measures the differential motherhood effect for top performers relative to non-top performers. Accordingly, the coefficient on

Table A13—THE MOTHERHOOD PENALTY AND ITS RELATION TO PERFORMANCE AROUND CHILDBIRTH

| | **Extensive margin:** Industry participation | # Leadership positions | Normalized AUM | **Intensive margin:** # Management responsibilities | P(single management) | P(active management) |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| **Panel A: Co-manager control group** | | | | | | |
| Motherhood | -0.308*** | -0.090** | -1.191*** | -0.831*** | -0.209*** | -0.272*** |
| | (0.022) | (0.040) | (0.353) | (0.153) | (0.035) | (0.015) |
| Motherhood × 3F CAR | -0.003 | -0.010 | -0.032 | 0.048 | 0.006 | -0.007 |
| | (0.005) | (0.010) | (0.043) | (0.036) | (0.006) | (0.006) |
| N | 39984 | 36830 | 36804 | 36804 | 36804 | 36804 |
| $R^2$ | 0.278 | 0.261 | 0.179 | 0.266 | 0.201 | 0.183 |
| **Panel B: Female co-manager control group** | | | | | | |
| Motherhood | -0.329*** | -0.057 | -1.155*** | -0.925*** | -0.221*** | -0.303*** |
| | (0.022) | (0.037) | (0.392) | (0.182) | (0.037) | (0.018) |
| Motherhood × 3F CAR | -0.003 | -0.006 | 0.008 | 0.053 | 0.009 | -0.008 |
| | (0.005) | (0.007) | (0.045) | (0.037) | (0.006) | (0.006) |
| N | 25585 | 22431 | 22405 | 22405 | 22405 | 22405 |
| $R^2$ | 0.247 | 0.295 | 0.208 | 0.290 | 0.235 | 0.218 |
| **Panel C: Female control group** | | | | | | |
| Motherhood | -0.300*** | -0.054* | -0.879*** | -0.750*** | -0.179*** | -0.250*** |
| | (0.019) | (0.032) | (0.288) | (0.131) | (0.033) | (0.016) |
| Motherhood × 3F CAR | -0.003 | -0.008 | 0.009 | 0.055 | 0.009* | -0.008 |
| | (0.005) | (0.007) | (0.049) | (0.037) | (0.005) | (0.006) |
| N | 64088 | 60934 | 60908 | 60908 | 60908 | 60908 |
| $R^2$ | 0.302 | 0.257 | 0.116 | 0.181 | 0.134 | 0.103 |
| Year-by-Cohort FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Age FE | Yes | Yes | Yes | Yes | Yes | Yes |

*Notes*: This table examines whether the motherhood penalty in career outcomes is related to managers' performance around childbirth. The coefficients represent difference-in-differences estimates based on Equation (1), allowing the motherhood penalty to differ by performance. Columns (1) to (3) focus on three key career outcomes: industry participation, the number of leadership positions, and person-level normalized AUM. Columns (4) to (6) report results for

## 5.8 Appendix robustness panels: selection, parallel trends, flows, and trading

**Read:**

- Table A3 finds no evidence that pregnancy timing is predicted by past performance or fund flows.
- Table A4 shows returners and non-returners are broadly similar before leave; if anything, returners look slightly better qualified.
- Table A6 supports the parallel-trends assumption for performance outcomes.
- Table A8 shows no systematic decline in institutional holdings or fund flows.
- Table A11 shows no broad change in trading behavior; total risk is stable, and non-market variation changes are modest and inconsistent.

**Economic message:** The appendix is important because it closes alternative explanations. The authors are not merely showing a career penalty; they show that common confounds - timing, attrition, investor punishment, and trading changes - do not account for the result.

Table A3—Is Pregnancy Timing Related to Past Performance or Fund Flows?

| | Pregnancy Next Month | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Past 12M 1F CAR | 0.000 | | |
| | (0.001) | | |
| Past 12M 3F CAR | | 0.000 | |
| | | (0.001) | |
| Past 4Q Cumulative Flows | | | -0.001 |
| | | | (0.003) |
| Manager Age | 0.002 | 0.002 | 0.001 |
| | (0.001) | (0.001) | (0.001) |
| PhD as highest degree | 0.028 | 0.027 | 0.023 |
| | (0.025) | (0.026) | (0.023) |
| Master as highest degree | 0.018 | 0.017 | 0.021 |
| | (0.019) | (0.019) | (0.019) |
| Fund Age (average) | 0.003 | 0.003 | 0.002 |
| | (0.002) | (0.002) | (0.002) |

**(a)** Table A3: pregnancy timing and past performance/flows

Table A4—Manager Characteristics at the Time of Maternity Leave: Returners vs. Non-Returners

| | Return date (N = 219) | No return date (N = 24) | Difference | ∣t-stat∣ |
|---|---|---|---|---|
| Manager age (in years) | 34.018 | 33.160 | 0.858* | 1.70 |
| Manager tenure (in years) | 1.625 | 1.583 | 0.041 | 0.19 |
| PhD as highest degree | 0.069 | 0.000 | 0.069*** | 4.00 |
| Master as highest degree | 0.899 | 1.000 | -0.101*** | 4.93 |
| Bachelor as highest degree | 0.032 | 0.000 | 0.032*** | 2.68 |
| # (Co-) managed funds | 3.677 | 3.120 | 0.557 | 0.99 |
| # Management responsibilities | 2.502 | 2.260 | 0.242 | 0.55 |
| # Leadership positions | 0.193 | 0.240 | -0.047 | 0.38 |
| Normalized AUM | 2.534 | 2.117 | 0.417 | 0.54 |
| P(Single management) | 0.606 | 0.640 | -0.034 | 0.34 |
| P(Active management) | 0.968 | 1.000 | -0.032*** | 2.68 |
| Pregnancy 3F Return (mean, in %) | 0.167 | -0.054 | 0.221 | 1.29 |
| Pregnancy 1F Return (mean, in %) | 0.120 | -0.064 | 0.185 | 0.99 |
| Pregnancy Raw Return (mean, in %) | 0.297 | 0.055 | 0.242 | 0.92 |

*Notes*: This table reports summary statistics for female fund managers at the time they begin maternity leave, distinguishing between managers who subsequently return to fund management (*Return*) and those for whom no return date is observed within the sample period (*No return*). The analysis is conducted at the manager–month corresponding to the start of maternity leave, identified using the RESSET Survivor-Bias-Free Mutual Fund database. The table reports means for returners and non-returners, the difference in means, and the associated *t*-statistics. Significance is calculated based on a two-sided t-test with standard errors clustered at the manager level. *, **, and ***, indicate statistical significance at the 10%, 5%, and 1% level, respectively. Variable definitions are provided in Appendix Table A1.

**(b)** Table A4: returners versus non-returners

Table A6—Testing the Parallel Trends Assumption

| | Treated | | Control | | Difference | |
|---|---|---|---|---|---|---|
| | Mean | N | Mean | N | Mean | t-statistic |
| **Panel A: Raw Return (in %)** | | | | | | |
| $-36 \leq t_P < 0$ | 0.405 | 8566 | 0.448 | 20446 | -0.042 | -0.89 |
| $-30 \leq t_P < 0$ | 0.422 | 8031 | 0.428 | 19067 | -0.006 | -0.12 |
| $-24 \leq t_P < 0$ | 0.447 | 7265 | 0.435 | 17119 | 0.012 | 0.20 |
| $-18 \leq t_P < 0$ | 0.444 | 6198 | 0.400 | 14472 | 0.044 | 0.72 |
| $-12 \leq t_P < 0$ | 0.452 | 4747 | 0.374 | 10943 | 0.079 | 1.18 |
| $-6 \leq t_P < 0$ | 0.378 | 2786 | 0.337 | 6220 | 0.041 | 0.47 |
| **Panel A: 1F Return (in %)** | | | | | | |
| $-36 \leq t_P < 0$ | 0.248 | 8429 | 0.254 | 20147 | -0.007 | -0.19 |
| $-30 \leq t_P < 0$ | 0.256 | 7905 | 0.251 | 18794 | 0.005 | 0.14 |
| $-24 \leq t_P < 0$ | 0.265 | 7153 | 0.250 | 16889 | 0.015 | 0.36 |
| $-18 \leq t_P < 0$ | 0.253 | 6106 | 0.231 | 14291 | 0.022 | 0.50 |
| $-12 \leq t_P < 0$ | 0.238 | 4676 | 0.238 | 10809 | -0.001 | -0.01 |
| $-6 \leq t_P < 0$ | 0.208 | 2745 | 0.207 | 6146 | 0.001 | 0.02 |
| **Panel B: 3F Return (in %)** | | | | | | |
| $-36 \leq t_P < 0$ | 0.246 | 8366 | 0.271 | 20015 | -0.065 | -1.41 |
| $-30 \leq t_P < 0$ | 0.365 | 5994 | 0.420 | 14083 | -0.054 | -1.12 |
| $-24 \leq t_P < 0$ | 0.390 | 5252 | 0.428 | 12223 | -0.037 | -0.71 |
| $-18 \leq t_P < 0$ | 0.402 | 4357 | 0.434 | 10072 | -0.032 | -0.56 |
| $-12 \leq t_P < 0$ | 0.402 | 3341 | 0.448 | 7601 | -0.046 | -0.74 |
| $-6 \leq t_P < 0$ | 0.394 | 1937 | 0.421 | 4258 | -0.027 | -0.36 |

*Notes*: This table reports results of tests for differences in means for monthly raw, 1F, and 3F abnormal returns between treated funds and non-treated matched control funds prior to pregnancy. Standard errors are clustered at the fund level. *, **, and ***, indicate statistical significance at the 10%, 5%, and 1% level, respectively.

**(a)** Table A6: parallel-trends checks

Table A8—Institutional Fund Holdings and Fund Flows around Childbirth

| | Institutional Holdings | | Flows | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Pregnancy ($0 \leq t_P < 9$) × Treated | -0.621 | -1.003 | 0.034 | 0.032 |
| | (3.129) | (2.932) | (0.049) | (0.049) |
| Post Maternity Leave ($0 < t_{PM} \leq 6$) × Treated | 1.781 | 1.072 | 0.057 | 0.055 |
| | (2.698) | (2.593) | (0.062) | (0.062) |
| Post Maternity Leave ($6 < t_{PM} \leq 12$) × Treated | 4.296 | 3.179 | 0.021 | 0.008 |
| | (3.036) | (2.979) | (0.072) | (0.072) |
| Post Maternity Leave ($12 < t_{PM} \leq 36$) × Treated | 6.313** | 5.654* | -0.004 | -0.010 |
| | (2.831) | (2.944) | (0.049) | (0.048) |
| Treated | -1.823 | -1.405 | -0.043 | -0.040 |
| | (2.035) | (1.986) | (0.035) | (0.034) |
| Log(Fund Size) | 4.610*** | 4.286*** | 0.059*** | 0.058*** |
| | (0.452) | (0.445) | (0.007) | (0.007) |
| Fund Age | -1.643*** | -1.714*** | -0.002 | -0.002 |
| | (0.484) | (0.486) | (0.002) | (0.002) |
| Manager Age | -0.224 | -0.131 | -0.004*** | -0.004*** |
| | (0.143) | (0.136) | (0.001) | (0.001) |
| PhD as highest degree | 0.351 | 0.039 | -0.056 | -0.059 |
| | (4.296) | (4.227) | (0.054) | (0.054) |
| Master as highest degree | 1.512 | 2.348 | -0.024 | -0.028 |
| | (3.167) | (3.134) | (0.045) | (0.045) |
| Month FE | Yes | No | Yes | No |
| Fund Type FE | Yes | No | Yes | No |
| Month × Fund Type FE | No | Yes | No | Yes |
| N | 17036 | 17036 | 39025 | 39025 |
| $R^2$ | 0.277 | 0.301 | 0.029 | 0.030 |

*Notes*: This table reports estimates from pooled panel regressions examining how institutional investors adjust their holdings and how fund flows change around childbirth. The sample includes all treated female managers and a control group consisting of all other (untreated) female managers. Columns (1) and (2) examine *Institutional Holdings*, measured as the share of fund assets held by institutional investors. These data are observed semi-annually (June and December), and only these months enter the regression. Columns (3) and (4) examine *Flows*, defined as quarterly fund flows; the analysis is restricted to observations from the first quarter of each year. The key explanatory variables are interactions of a treatment indicator with indicator variables for *Pregnancy* ($0 \leq t_P < 9$) and three mutually exclusive post-maternity-leave periods: *Post Maternity Leave* ($0 < t_{PM} \leq 6$), *Post Maternity Leave* ($6 < t_{PM} \leq 12$), and *Post Maternity Leave* ($12 < t_{PM} \leq 36$). *Before* (the 36 months prior to pregnancy) is the omitted category. Control variables are the same as in Table 3. All regressions include month fixed effects; some specifications additionally include manager and/or fund fixed effects, as indicated in the table. Standard errors are clustered at the fund and month level. *, **, and ***, indicate significance at the 10%, 5%, and 1% levels, respectively.

**(b)** Table A8: institutional holdings and flows

Table A11—TRADING BEHAVIOR AROUND CHILDBIRTH

| | Total Risk (in %) | | Non-Market Variation | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Pregnancy ($0 \leq t_P < 9$) × Treated | -0.079 | -0.057 | 0.010 | 0.020* |
| | (0.091) | (0.085) | (0.012) | (0.011) |
| Post Maternity Leave ($0 < t_{PM} \leq 6$) × Treated | -0.037 | -0.006 | 0.021 | 0.027* |
| | (0.116) | (0.112) | (0.015) | (0.014) |
| Post Maternity Leave ($6 < t_{PM} \leq 12$) × Treated | 0.067 | 0.068 | 0.013 | 0.017 |
| | (0.134) | (0.127) | (0.016) | (0.015) |
| Post Maternity Leave ($12 < t_{PM} \leq 36$) × Treated | -0.069 | -0.037 | 0.011 | 0.020 |
| | (0.157) | (0.156) | (0.017) | (0.016) |
| Pregnancy ($0 \leq t_P < 9$) × Treated × Team Fund | 0.155 | 0.130 | -0.007 | -0.015 |
| | (0.149) | (0.141) | (0.017) | (0.016) |
| Post Maternity Leave ($0 < t_{PM} \leq 6$) × Treated × Team Fund | 0.111 | 0.062 | -0.019 | -0.022 |
| | (0.196) | (0.192) | (0.021) | (0.021) |
| Post Maternity Leave ($6 < t_{PM} \leq 12$) × Treated × Team Fund | -0.136 | -0.134 | -0.011 | -0.019 |
| | (0.214) | (0.209) | (0.024) | (0.023) |
| Post Maternity Leave ($12 < t_{PM} \leq 36$) × Treated × Team Fund | -0.021 | -0.032 | -0.011 | -0.018 |
| | (0.246) | (0.245) | (0.025) | (0.025) |
| Treated | 0.051 | 0.001 | -0.018 | -0.025** |
| | (0.102) | (0.099) | (0.013) | (0.012) |
| Team Fund | -0.493*** | -0.475*** | 0.007 | 0.012 |
| | (0.174) | (0.170) | (0.017) | (0.017) |
| Month FE | Yes | No | Yes | No |
| Fund Type FE | Yes | No | Yes | No |
| Month × Type FE | No | Yes | No | Yes |
| N | 107839 | 107815 | 97243 | 97241 |
| $R^2$ | 0.549 | 0.586 | 0.213 | 0.289 |

*Notes*: This table reports coefficients from pooled panel regressions of fund-level trading behavior around childbirth that include all treated female managers and a control group consisting of all other (untreated) female managers. *Total Risk* is the annualized standard deviation of fund returns within a month, computed by multiplying the within-month standard deviation by $\sqrt{252/12}$ and expressed in percent. *Non-Market Variation* is defined as one minus the $R^2$ from a regression of fund excess returns on the combined market factor model within a month and captures the share of fund return variation not explained by market movements; it is expressed in percent. The key explanatory variables include (i) interactions of the treatment indicator with event-time period indicators, and (ii) triple interactions that additionally allow the event-time effects to differ between single-managed and team-managed funds for treated managers. For untreated controls, the team indicator is set to zero such that the single versus team decomposition is identified off variation among treated managers only (consistent with the main specification). *Before* denotes the 36 months prior to pregnancy and serves as the omitted category. *Pregnancy* ($0 \leq t_P < 9$) denotes the months between becoming pregnant and the start of maternity leave. *Post Maternity Leave* ($0 < t_{PM} \leq 6$), *Post Maternity Leave* ($6 < t_{PM} \leq 12$), and *Post Maternity Leave* ($12 < t_{PM} \leq 36$) denote the corresponding post-leave subperiods. Control variables are the same as in Table 3 and described in Appendix Table A1. Standard errors are clustered at the fund and month levels. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A11: trading behavior around childbirth

# 6 Short summary

- The paper studies motherhood penalties among Chinese mutual fund managers.
- The setting is powerful because maternity leaves are disclosed and performance is observable.
- The final maternity-leave sample contains 243 unique managers after excluding repeated maternity leaves.
- The main comparison group is the mother's pre-birth co-managers on the same funds.
- Event studies show clean pre-trends and sharp post-birth declines in career outcomes.
- New mothers are 25.5 percentage points less likely to remain active in the industry, about a 42% reduction relative to the mean.
- Leadership roles, normalized AUM, management responsibilities, single-manager mandates, and active-management mandates all decline.
- Performance falls temporarily during pregnancy for single-managed funds but recovers after leave.
- Team-managed funds show weaker performance effects, suggesting co-managers buffer temporary disruptions.
- Fund flows and institutional holdings do not decline around childbirth.
- Managers substitute away from on-site visits, but online visits and trading behavior remain broadly stable.
- Fertility timing is not predicted by past performance or flows.

- Returners and non-returners look broadly similar before leave.
- Top performers do not experience smaller motherhood penalties.
- The evidence points toward post-leave retention and responsibility allocation rather than sustained productivity decline.

# 7 Conclusion

## 7.1 Core conclusion

Ginzinger, Li, Niessen-Ruenzi, and Wang show that motherhood generates large and persistent career penalties in the Chinese mutual fund industry. New mothers are less likely to remain active, and those who stay receive fewer leadership roles, less normalized AUM, and fewer prestigious management responsibilities. These penalties are robust to comparisons with co-managers, female co-managers, and other non-mother female managers.

## 7.2 Economic intuition

The central intuition is that the paper finds a mismatch between observable productivity and career allocation. In mutual fund management, productivity is unusually measurable through returns, abnormal returns, fund flows, AUM, investor response, and information acquisition. If the motherhood penalty were mainly productivity-based, mothers should experience persistent underperformance, investor withdrawals, and career losses that are closely tied to poor fund outcomes.

The evidence does not fit that story. Performance declines are temporary, mainly around pregnancy, and largely recover after maternity leave. Investor flows do not fall sharply, and high-performing mothers are not protected from career penalties. Pregnancy-period performance also does not explain later career losses.

The deeper mechanism is therefore likely about the allocation of career capital. Maternity leave can trigger a temporary reassignment of funds, clients, leadership roles, or single-manager mandates. Once these responsibilities are shifted to others, they may not fully return, even if the mother remains productive.

$$\text{Maternity leave} \Rightarrow \text{temporary role reallocation} \Rightarrow \text{loss of career capital} \Rightarrow \text{persistent career penalty.}$$

Thus, the motherhood penalty is not mainly that mothers become worse investors. It is that childbirth changes how organizations allocate scarce career opportunities such as AUM, leadership, active mandates, and portfolio responsibility.

## 7.3 Incremental contribution of the paper

### 7.3.1 Core incremental contribution

The paper's incremental contribution is not simply that it documents a motherhood penalty. A large labor-economics literature already shows that childbirth is associated with lower female labor-force attachment, earnings, promotion, and career progression. The new contribution is that this paper studies the motherhood penalty in a setting where worker productivity is unusually observable, financially important, and measured at high frequency: the mutual fund industry.

This setting makes the paper a particularly sharp test of performance-based explanations for the motherhood penalty. In many occupations, it is difficult to know whether mothers lose career opportunities because their productivity falls, because employers discriminate, because clients react negatively, or because the worker voluntarily reduces effort. In fund management, however, the paper can observe returns, abnormal performance, fund flows, assets under management, leadership roles, portfolio responsibility, corporate site visits, and trading behavior. This lets the authors ask a much deeper question:

Is the motherhood penalty proportional to observable productivity changes?

The answer is largely no. Career penalties are large and persistent, while observable performance effects are temporary, concentrated mainly during pregnancy for single-managed funds, and do not predict later career losses. This mismatch between career outcomes and measured productivity is the paper's central incremental insight.

### 7.3.2 Contribution relative to the motherhood-penalty literature

The first contribution is to move the motherhood-penalty literature into a high-skill, high-pay, output-measurable finance occupation. Much existing work studies wages, labor supply, hours worked, or broad occupational outcomes. This paper instead studies a setting where performance is public, market-valued, and continuously updated.

This matters because finance is a difficult environment for purely productivity-based stories. If a fund manager performs poorly, the market can observe poor returns and investors can withdraw capital. If a fund manager performs well, that performance is also visible. Therefore, if mothers lose responsibility even when performance recovers and investors do not punish them, the result is more informative than in occupations where output is hard to measure.

**Incremental message.** The paper shows that even in a highly measurable occupation, childbirth can produce large career penalties that are not well explained by observed productivity. This strengthens the argument that motherhood penalties are not only about unmeasured output loss.

### 7.3.3 Contribution relative to finance research

The second contribution is to bring family and gender frictions into the study of asset-management labor markets. The mutual fund literature usually studies performance, flows, incentives, manager skill, fund families, and portfolio choice. This paper adds a new labor-market dimension: how a major life event affects the allocation of assets, leadership, and responsibility inside fund companies.

The paper shows that childbirth affects not only whether women remain active in the industry, but also the type and quality of responsibilities they receive after returning. The decline in normalized AUM, single-manager mandates, active-management mandates, and leadership positions suggests that the motherhood penalty operates through the internal allocation of career opportunities, not just through formal employment status.

**Incremental message.** The paper treats AUM and portfolio responsibility as career capital. Losing AUM is similar to losing promotion prospects, compensation potential, investor visibility, and organizational influence.

### 7.3.4 Contribution of the maternity-leave disclosure design

A third contribution is measurement. In most datasets, childbirth and maternity leave are hard to observe cleanly, especially for elite professional workers. This paper exploits the regulatory disclosure rule in the Chinese mutual fund industry: fund manager absences exceeding 30 days must be publicly disclosed. Since maternity leave exceeds this threshold, the authors can identify maternity leave events, approximate pregnancy timing, return dates, and post-return careers.

This creates a rare event-study setting in which the researchers observe:

- the timing of pregnancy and maternity leave;
- whether the manager returns;

- the manager's pre- and post-birth fund responsibilities;
- fund performance during pregnancy, leave, and after return;
- investor flows and institutional ownership;
- effort proxies such as corporate site visits;
- comparisons with pre-birth co-managers.

**Incremental message.** The paper turns a usually private life event into a precisely dated career event, which allows a much cleaner analysis of the motherhood penalty than standard administrative or survey datasets.

#### 7.3.5 Contribution of the co-manager comparison

A fourth contribution is the co-manager design. Instead of comparing mothers to all other fund managers, the paper compares each new mother to her pre-birth co-managers on the same fund. This is a powerful counterfactual because co-managers share the same fund family, investment mandate, fund-level shocks, client base, and pre-event professional environment.

This comparison helps address the concern that mothers are different because they work at different firms, manage different funds, face different market conditions, or have different career tracks. The co-manager was working in the same fund environment immediately before the childbirth event.

$$\text{Mother's post-birth outcome} - \text{pre-birth co-manager's post-event outcome}$$

is therefore closer to a within-workplace comparison than a broad labor-market comparison.

**Incremental message.** The paper does not merely document a gender gap. It documents divergence between people who were professional peers and direct collaborators before childbirth.

#### 7.3.6 Contribution to mechanism: what the paper rules out

A fifth contribution is that the paper systematically tests several plausible explanations and shows that none fully explains the career penalty.

- **Persistent performance decline.** Returns fall temporarily during pregnancy for some single-managed funds, but performance rebounds after maternity leave.
- **Investor punishment.** Fund flows and institutional ownership do not systematically decline around childbirth.
- **Lower observable effort.** On-site visits decline, but online visits and total information acquisition are more stable, suggesting substitution rather than complete withdrawal.
- **Lower pre-birth skill.** Strong pre-birth performers do not receive a meaningfully smaller penalty.
- **Realized poor performance during pregnancy.** Pregnancy-period performance does not predict the later career penalty.

This is important because the paper's strongest contribution is not proving one exact mechanism. Rather, it narrows the set of plausible mechanisms. The evidence points away from simple productivity, investor-demand, and performance-selection explanations.

**Incremental message.** The paper shows that the motherhood penalty is not aligned with the main observable signals that normally determine careers in asset management.

### 7.3.7 Contribution to organizational economics

The paper also contributes to organizational economics by showing how temporary leave can become a persistent career shock. Maternity leave may initially require a temporary reallocation of clients, funds, portfolios, and leadership responsibilities. But once responsibilities are reassigned, they may not fully return. This creates organizational inertia.

The mechanism can be summarized as:

$$\begin{aligned} \text{childbirth} \Rightarrow \text{temporary leave} &\Rightarrow \text{temporary replacement or co-manager reallocation} \\ &\Rightarrow \text{persistent loss of AUM, mandates, and leadership opportunities.} \end{aligned}$$

This mechanism does not require investors to punish mothers or mothers to perform badly. It only requires that career opportunities are path dependent. In high-status occupations, even short interruptions can have long-run effects if promotions, client assignments, and responsibility allocation are tournament-like.

**Incremental message.** The paper suggests that motherhood penalties may arise through the internal reallocation of scarce career capital, not only through wage cuts or formal exits.

### 7.3.8 Why the paper is especially convincing

The paper is convincing because the main career result is paired with many mechanism tests. The authors do not stop at showing that mothers are less likely to stay active. They also ask whether mothers perform worse, whether investors withdraw money, whether effort falls, whether teams mitigate performance loss, whether strong performers are protected, and whether pregnancy performance predicts future penalties.

This layered structure makes the interpretation stronger. A single career regression could be consistent with many explanations. But the combination of career, performance, flow, effort, and heterogeneity evidence makes the productivity explanation much less persuasive.

**Insight.** The paper's contribution is strongest because of the gap between what is large and what is small:

$$\text{large persistent career penalty} \quad \text{versus} \quad \text{small temporary observable productivity effect.}$$

### 7.3.9 Limits of the contribution

The paper does not fully separate supply-side and demand-side mechanisms. Some mothers may voluntarily reduce responsibilities after childbirth, while fund companies may also reallocate opportunities away from them. The data cannot perfectly distinguish preferences, family constraints, employer beliefs, and discrimination.

However, this limitation does not weaken the central contribution. Even if some of the career penalty reflects choice, the paper still shows that the penalty is not closely tied to observed fund performance or investor response. The key result is that childbirth changes career trajectories in a way that is much larger and more persistent than the observed performance shock.

## 7.4 Takeaway

The paper's broader insight is that high-powered, performance-measured labor markets can still generate large motherhood penalties. Observable performance does not automatically protect women from career setbacks after childbirth. In finance, the crucial inequality margin may be less about realized investment skill and more about who gets responsibility, assets, leadership, and second chances after a major life event.